

3.2.6. Numpy: Searching, Sorting, Counting, Broadcasting

07:39

The given code in the editor takes a single array, `array1`, as space-separated integers as input from the user.

Additionally, it takes the following inputs:

- `search_value`: The value to search for in the array.
- `count_value`: The value to count its occurrences in the array.
- `broadcast_value`: The value to add for broadcasting across the array.

You need to complete the code to perform the following operations:

- 1. Searching:** Find the indices where `search_value` appears in `array1` and print these indices.
- 2. Counting:** Count how many times `count_value` appears in `array1` and print the count.
- 3. Broadcasting:** Add `broadcast_value` to each element of `array1` using broadcasting, and print the resulting array.
- 4. Sorting:** Sort `array1` in ascending order and print the sorted array.

Input Format:

1. A single line containing space-separated integers representing `array1`.
2. An integer `search_value` represents the value to search for in the array.
3. An integer `count_value` represents the value to count in the array.
4. An integer `broadcast_value` represents the value to add to each element of the array.

Output Format:

1. The indices where `search_value` occurs in `array1`.
2. The count of occurrences of `count_value` in `array1`.
3. The array after adding the `broadcast_value` to each element.
4. The sorted array.

Code:

```
import numpy as np
```

```
# Input array from the user
```

```
array1 = np.array(list(map(int, input().split())))
```

```
# Searching
```

```
search_value = int(input("Value to search: "))
```

```
count_value = int(input("Value to count: "))
```

```
broadcast_value = int(input("Value to add: "))
```

```
# 1. Find indices where value matches in array1
```

```
search_indices = np.where(array1 == search_value)[0] # np.where returns a tuple, we take the first element
```

```
print(search_indices)
```

```
# 2. Count occurrences in array1
```

```
count_occurrences = np.count_nonzero(array1 == count_value)
```

```
print(count_occurrences)
```

```
# 3. Broadcasting addition (Add broadcast_value to each element)
```

```
broadcasted_array = array1 + broadcast_value
```

```
print(broadcasted_array)
```

```
# 4. Sort the array
```

```
sorted_array = np.sort(array1)
```

```
print(sorted_array)
```

Average time

0.059 s

58.75 ms



Maximum time

0.069 s

69.00 ms



✓ 2 out of 2 shown test case(s) passed

✓ 2 out of 2 hidden test case(s) passed

✓ Test case 1 68 ms

Debug



Expected output

1 1 1 2 2 2

Value to search: 1

Value to count: 2

Value to add: 2

[0, 1, 2]

3

[3, 3, 3, 4, 4, 4]

[1, 1, 1, 2, 2, 2]

Actual output

1 1 1 2 2 2

Value to search: 1

Value to count: 2

Value to add: 2

[0, 1, 2]

3

[3, 3, 3, 4, 4, 4]

[1, 1, 1, 2, 2, 2]

✓ Test case 2 50 ms

Debug



Expected output

1 2 3 4 5

Value to search: 6

Value to count: 6

Value to add: 6

[]

0

[, 7, 8, 9, 10, 11]

[1, 2, 3, 4, 5]

Actual output

1 2 3 4 5

Value to search: 6

Value to count: 6

Value to add: 6

[]

0

[, 7, 8, 9, 10, 11]

[1, 2, 3, 4, 5]